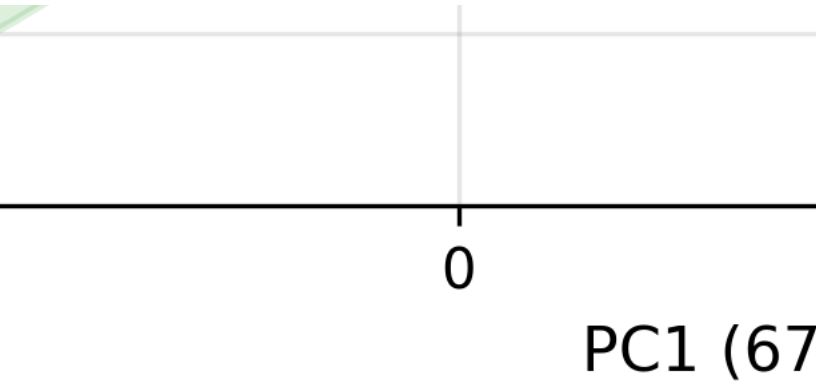


$\times^{TM} x \text{---} x^{TM} x'' x'' x\check{z} \times \square x' x'' x^a \times \square x^a x\check{z} x'' x' \times^{TM} x^a x'' x \square x'' x^{TM} x \sim x^{TM} x \square, x'' x\check{z} x^a \times \square x^{TM} x\check{z} x'' x$
 $\times^{TM} x^a \times \S x'' x x'' x^{TM} x \bullet x^a x'' x x'' \times \square x^{TM} x^a x' \times^{TM} x \text{---} x; x^{TM} Nb/Zr x \times \check{z} x \bullet x \times x^{TM} x \square x^{TM} x \bullet x^a x''$,
 $x' x' x \phi x'' x \S Zr/Nb x' x' x \bullet x'' x\check{z} x \square x \bullet x'' x'' x\check{z} x^a \times \square x^{TM} x \square \times \ae x \check{z} \times \S x \bullet x'' Bing\tilde{A} \P l A x'' x \square x'' x''$ -
 $.^{TM} x x^{TM} x \times \square \times \ae x \S x \square \times \ae$

$\times^{TM} x \check{z} \times \ae x^{TM} x \odot \times \ae k=3 \times \S x' x \bullet x_j \times \bullet x^a, x'' x^a \times \bullet x \square \times \square x' \times \square \times \bullet x \square \times \check{Y} \times \times \ae \times \ae x^{TM} x \square x^a \times^{TM} x \times$
 $\times \bullet x' x'' x\check{z} x'' (x \square \times^{TM} x \bullet x'' 6) x\check{z} x_j \times^{TM} x' x'' x' \times \square \times \bullet x \square \times \check{Y} x'' x \bullet x \check{z} x'' x \S x' x \bullet x_j x'' x\check{z} x'' x \bullet x'' \times \S x^a$
(), $\times \times \odot x \text{---} x \square \times \P Bing\tilde{A} \P l A \times \check{z} x; x^a \times \phi \times f \times \check{z} \times \check{z} x x'' x' x' x \bullet x' x'' x\check{z} x^{TM} x \text{---} x \bullet x' x' x'' \times \bullet x \ae \times \check{z}$

A scatter plot showing the relationship between Nb (x-axis) and Rb (y-axis). The data points are colored green, orange, and blue. A green shaded region is visible in the upper right quadrant. Three arrows point from a central cluster of points to the labels 'Zr', 'Nb', and 'Rb'.



uted from element ratios Rb/Zr, Nb/Zr, Rb/Nb.

.Figure 5: PCA of all attributed obsidian items, coloured by site

on standardised element ratios).

Dendrogram. The single outlier at high distance (yif_10671, BingĀ¶l A) is visible at the top right

“ × • × § × Ž × ” ×_i × • × — ×TM ×TM × □ × • × < ×TM ×_i ×

39.7%) × " ×TM × • × ' × ϕ × œ ×TM ×TM ×TM × — × • ×_i × Ž × " ×TM × Ž × Ÿ (× ' ×^a × • × § × □ × œ ×TM × □ ×_i ×^a ×_i ×
(. 477 (94.6%) × Ž × " × □ × " ×TM × " ×TM × □ × " ×_i ×TM × ' × • × " ×_i ×) × Ž × " × ' ×TM × Ÿ ×TM ×TM × — × • ×_i 2-
× • ×^a × Ž × © × ×TM, × " × Ž × □ × © × " ×^a × □ ×TM ×^a × × • ×^a × " × ' ×TM × © × " × Ž × ' × • ×_i ×_i ×^a Zr/Nb,
. × œ × Ž × " × • ×^a × § ×TM × — × • × — ×) ×TM × • × œ × " - Rb

). GolluDag and EGD together represent the GĀ¶llĀ¼ DaĀŸ East complex (see Section 5.3 for
.explanation of the split)

olluDag EGD BingolA Total

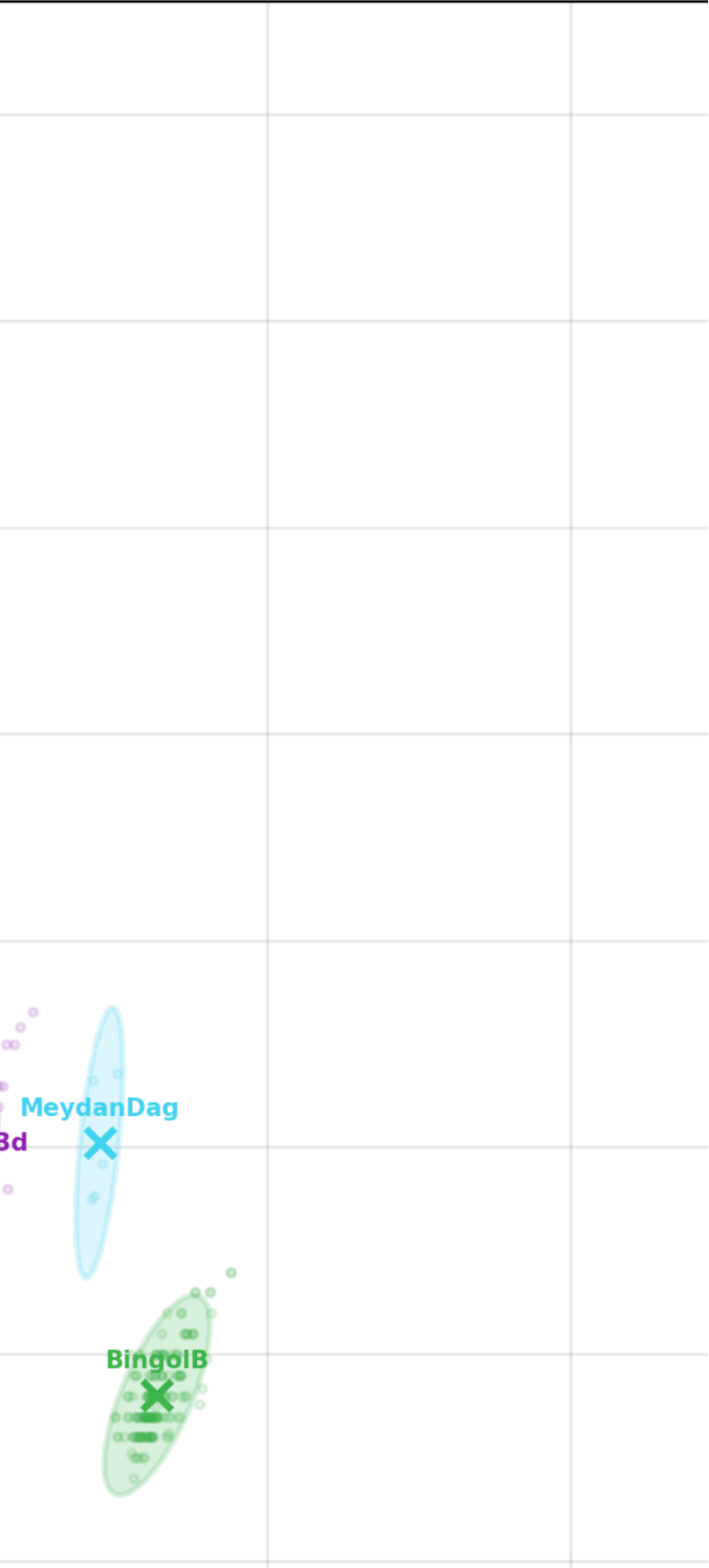
50	0	379
11	0	103
1	1	22
62	1	504

× — ×_i × • × œ - GolluDag × • - 50 (13.2%) × œ - EGD. GĀ¶llĀ¼ DaĀŸ East × © × • × œ × " ×^a × œ × œ × □ ×
. □ ×_i × • × ϕ × • × " × " ×TM × Ÿ × œ × □ × • × " × § × " ×_i × £ × " - EPPNB × ' × Ž

TM × " ×TM × □ (89.3%) ×TM × • × — ×_i × • × œ - GolluDag × • - 11 (10.7%) × œ - EGD. × " × □ × • ×_i × ×TM × ϕ ×
© × œ × Ž × • ×_i × □, × Ž × " × © × Ž ×_i × ' ×TM × ϕ × ϕ × œ × □ × • ×^a × " × " × © ×^a × □ ×_i × □ × § × " × □ × ϕ ×TM × œ × "
. × ' × ' × œ × œ × □ × • × " ×) × • × © × œ × Ž × ϕ × ' × " × " × " × • × □ ×TM × □ × Ÿ â € “ PPNB

œ - GolluDag, 1 × œ - EGD, 1 × œ - Group3d (× œ × □ × • × " × " × □ ×TM; × " × □ × • ×_i × ϕ ×TM × £ 5.5), × □ ×TM ×
. × • - 1 × œ - BingolA (× " × □ × • ×_i × ϕ ×TM × £ 5.4)

Source a
(shaded = 95% Maha



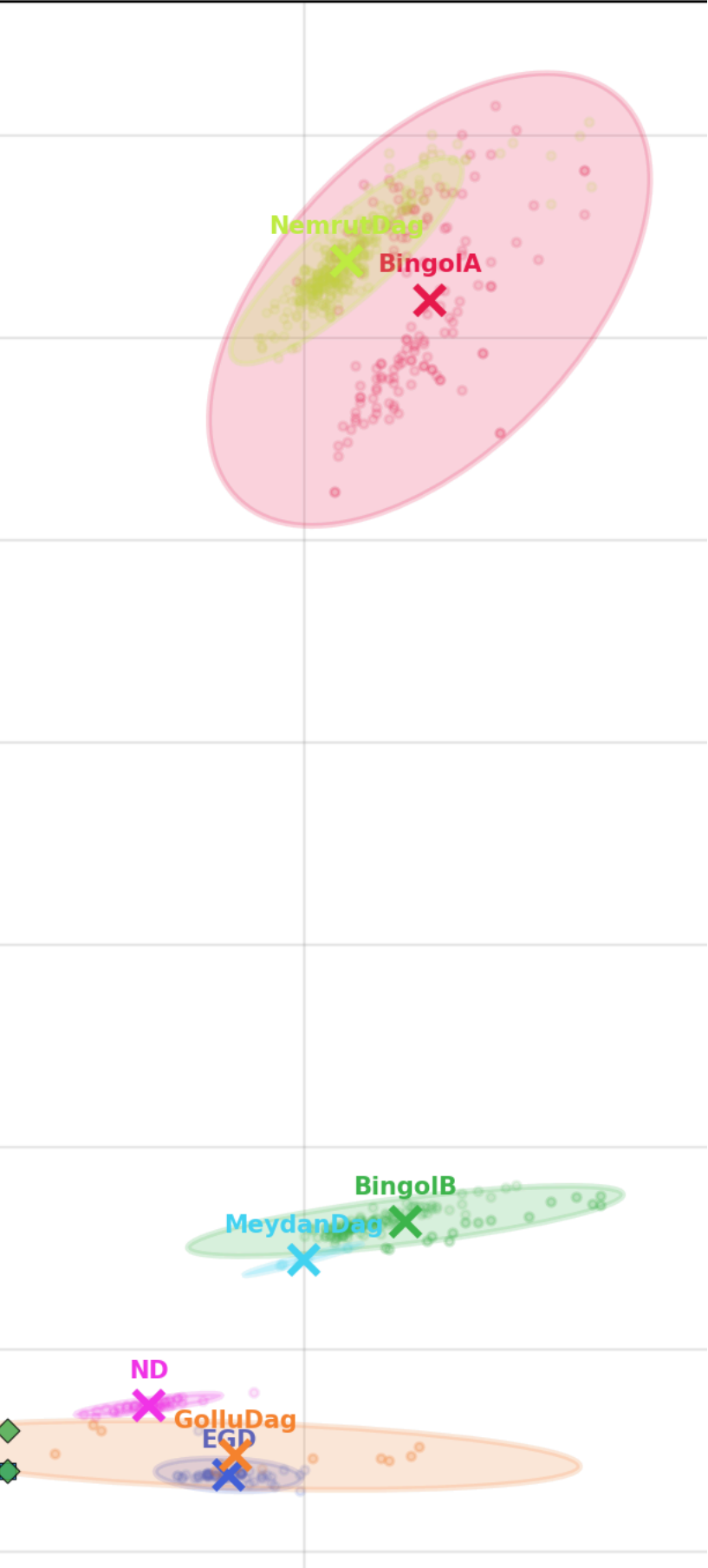
400

600

ier-1 reference sources and our attributed sample points. Site symbols: squares = Motza,

arily with the GolluDag and EGD ellipses (GÄ¶llÄ¼ DaŸŸ East complex). yif_10671 (BingÄ¶l
.A) appears in the upper right at Zr â%o^ 1005 ppm, off the plot edge if not log-scaled

Rb vs Zr w
(Note: sample Rb is ~h



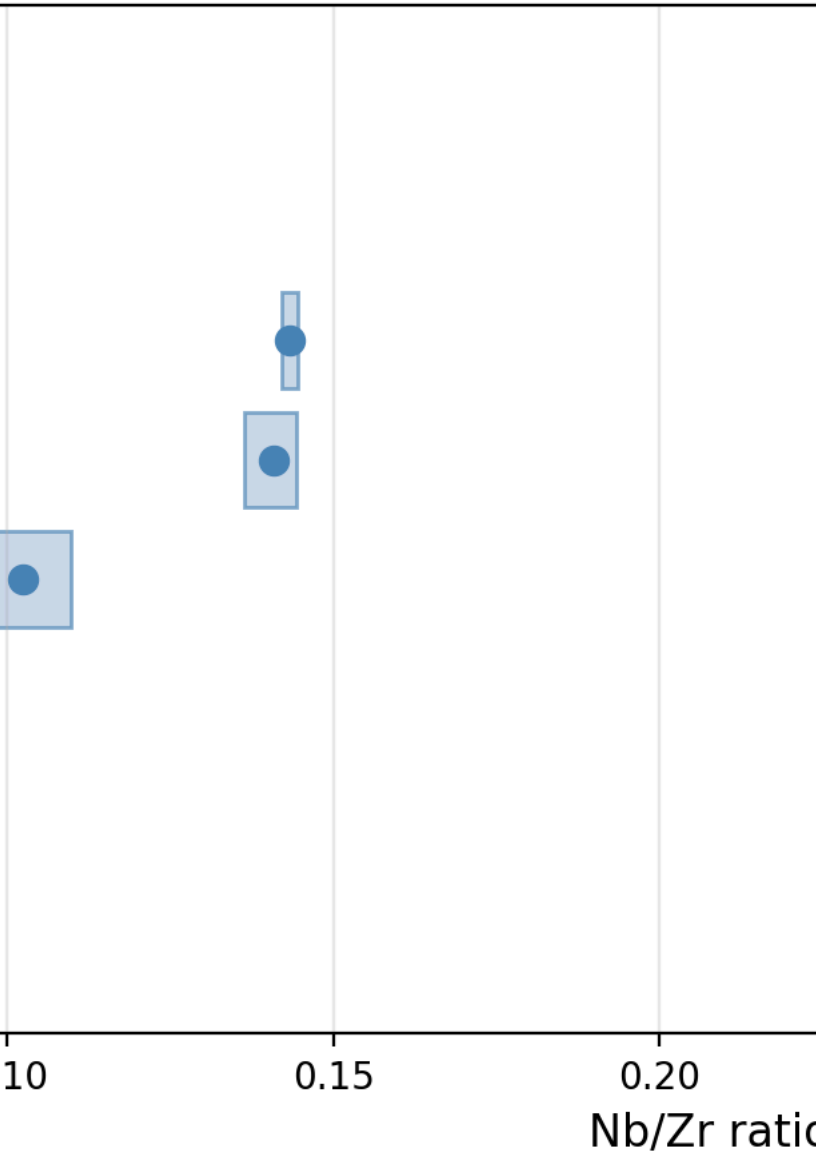
200

Rb (ppm) [sample

relative to reference source ellipses.

to the $G\ddot{A}rll\ddot{A}Da\ddot{Y}$ reference ellipse, consistent with the Rb calibration offset described in Section 4.2.

Nb/Zr: reference sources vs
(IQR boxes = source geochemical range)



ians.

edians coincide with the GolluDag reference IQR, confirming the dominant source assignment

S



$\times \times \times \times \hat{\mathbf{a}}\mathbf{c}^{\mathbf{T}}$ **GolluDag** $\times \bullet \mathbf{EGD} \hat{\mathbf{a}}\mathbf{c}^{\mathbf{T}}$ $\times \hat{\mathbf{z}} \times^{\mathbf{d}} \times^{\mathbf{TM}} \times^{\mathbf{TM}} \times - \times \times \times^{\mathbf{d}} \times \times \times^{\mathbf{a}} \times^{\mathbf{TM}} \times \times \hat{\mathbf{Y}}$ $\times \mathbf{re} \mathbf{-GA} \mathbf{\Pi} \mathbf{\Lambda} \times \times \times$
 $\times \times \bullet \times \hat{\mathbf{s}} \times^{\mathbf{TM}} \times \times, \times \times \times \times \hat{\mathbf{s}} \times^{\mathbf{TM}} \times \times (\times) \mathbf{-38}^{\mathbf{d}} \mathbf{N}, \mathbf{34}^{\mathbf{d}} \mathbf{E}.$ $\times \times \hat{\mathbf{Y}} \times \hat{\mathbf{z}} \times \times \times \times^{\mathbf{TM}} \times \times \times^{\mathbf{d}} \times \times \times \times^{\mathbf{TM}} \times \times^{\mathbf{a}}$
 $: \square \times^{\mathbf{TM}} \times \times \times \times \times \square \times \square \times^{\mathbf{TM}} \times \times \times \times \times \times \times \times^{\mathbf{d}} \times \hat{\mathbf{z}} \times \times \times \times^{\mathbf{TM}} \times \times \times \hat{\mathbf{z}} \times \times \times \times \times \hat{\mathbf{Y}} \times \times \times \times \times \times \times \times \hat{\mathbf{z}}$

[illegible][illegible]

13/16

- k (Turkey) and its wider implications for Near Eastern trade. *Archaeometry* 48, 507â€“516
- .Carter, T., et al.Â (2008). The chipped stone assemblage from Basta. *Neo-Lithics* 2008
- n Kortik Tepe and Tell Aswad, Syria. *Journal of Archaeological Science* 40, 3804â€“3815
- m-curved obsidian from the Near East. *Journal of Archaeological Science* 39, 728â€“736
- RF for obsidian provenance analysis. *Journal of Archaeological Science* 40, 1080â€“1093
- with portable XRF: accuracy, precision, and field conditions. *Archaeometry* 56, 351â€“373
- idian from the Zagros region: a contribution to the study of prehistoric exchange. *Journal*
• *.of Archaeological Science: Reports* 11, 643â€“658
- idian excavated from Chalcolithic and Bronze Age levels at the sites of Tell Masaikh and
• .Qalâ€™at el-Mudiq, Syria. *Archaeometry* 51, 879â€“893
- early cultural contact in the Near East. *Proceedings of the Prehistoric Society* 32, 30â€“72
- e and Early Bronze Age assemblages in the Negev. *Journal of Archaeological Science* 38,
• .1062â€“1069
- e Chalcolithic southern Levant. *Journal of Archaeological Science: Reports* 8, 430â€“440
- le XRF in archaeological obsidian characterization studies? *Archaeometry* 52, 793â€“798
- kel, Y. (1986). Provenience of the Shaâ€™ar Hagolan obsidian. *Palâ€™orient* 12, 81â€“83
- obsidian from Tell es-Safi/ Gath, Israel. *Journal of Archaeological Science* 34, 905â€“913
- d neighboring countries during the fourth to second millennia B.C. *Archaeometry* 22, 110
- vation analysis of obsidian from Israel and the near east. *MASCA Journal* 1(7), 206â€“209